

Muditya Raghav

Software Engineer — Backend Systems, Full-Stack Products & Test Automation

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EDUCATION

Gyan Ganga Institute of Technology and Sciences

Aug. 2019 – May 2023

Bachelor of Technology

Jabalpur, India

EXPERIENCE

Software Engineer (Freelance)

April 2025 – Present

Personal Engineering Practice

Jabalpur, India

- Designed and shipped **three end-to-end software systems** — **Clara Network**, **Malcom**, and **Kingswork** — spanning payment infrastructure, AI-powered research, and trading intelligence
- Architected and deployed backend services using *Go*, *Python*, *FastAPI*, *REST APIs*, and *WebSockets*, with *Docker*, *AWS EC2*, *Linux*, and *CI/CD* for deployment and operations

QA Associate Engineer

Dec. 2023 – March 2025

Reliance Jio Platforms

Hyderabad, India

- Automated **80+ test cases** using *Automator AI* with LLMs, boosting test coverage by **40%**
- Enhanced REST API suites, fixing **35+ critical bugs** and improving system stability by **25%**
- Integrated *CI/CD* pipelines with *Jenkins*, accelerating releases by **20%**

Software Engineer Intern

April 2022 – June 2022

Persistent Systems

Remote

- Automated Excel-based QA workflows using *Python* and *Pandas*, reducing test time by **90%**; built a validation engine processing **10,000+ rows per test**
- Developed cross-functional scripts adopted by **2 internal teams**, eliminating redundant work

Cyber Security Intern

April 2021 – July 2021

Cisco Network

Remote

- Conducted penetration tests on **3+ live web platforms** using *Burp Suite* and *Nmap*, discovering **5+ critical bugs** and submitting **3 verified reports** to PayPal's Bug Bounty Program via *HackerOne*

PROJECTS

Clara Network | *Go*

Live | GitHub

- Designed and built a **Mastercard/Visa-style card payment network** end-to-end in *Go*: **ISO 8583** switch, BIN-based authorization routing with failover/stand-in processing, and an append-only double-entry ledger with reconciliation
- Implemented issuing (**EMV ARQC** verification, token vault, wallet provisioning) and acquiring (merchant boarding, **MATCH/OFAC** screening) stacks
- Built a disputes engine and an **HSM**-based key-management layer (dual-control ceremonies, AES key wrap, **ISO 9564/9797-1**) plus an instant-payments layer (**ISO 20022**) for **24/7 real-time settlement**

Malcom | *Next.js*, *TypeScript*, *Supabase*, *SQLite*, *Stripe*

Live | GitHub

- Built a **full-stack AI research platform** with streamed **LLM** responses, document context retrieval, web research with sources, and **Markdown**/math rendering
- Implemented saved chats, secure sessions, and **Stripe**-backed subscriptions on a *Next.js/Supabase* stack

Kingswork | *FastAPI*, *React*, *WebSockets*, *SQLAlchemy*

Live | GitHub

- Built a **full-stack trading intelligence platform** with dashboards, stock analysis, paper portfolios, backtesting, and AI-assisted research
- Developed a multi-domain *FastAPI* backend with **JWT** auth, *WebSockets*, and *SQLAlchemy* models

TECHNICAL SKILLS

Languages: Python, JavaScript/TypeScript, **Go**, SQL, C++ (familiar)

Frameworks & Tools: React, Next.js, Express, FastAPI, Selenium, Cypress, Playwright, Postman, Jenkins, Git

Data & Infra: SQLite, MongoDB, Supabase, Docker, Linux, AWS EC2, REST APIs, WebSockets, CI/CD

Security: Burp Suite, Nmap, DVWA, Juice Shop, Wireshark